# Machine Learning-Based Drug Repurposing Discovery for Non-Globular MYC and CTNNB1 Oncoproteins

Computer *Engineering Senior Design Project*

Author: Beyza Yörük

beyzayoruk@posta.mu.edu.tr

Advisor: Assoc. Prof. Dr. Tugba Önal SÜZEK

Muğla Sıtkı Koçman University

Department of Computer Engineering

# Defining the Oncogenic Targets

## MYC Proto-Oncogene

Target Name: MYC Proto-Oncogene

PDB ID: 6G6K

Biological Role: Transcription factor; overexpressed in ~70% of cancers

Raw Bioassay Files: ~40 CSV files (PubChem)

Curated Samples: 660 (73% active)

## Beta-Catenin (CTNNB1)

Target Name: Beta-Catenin (CTNNB1)

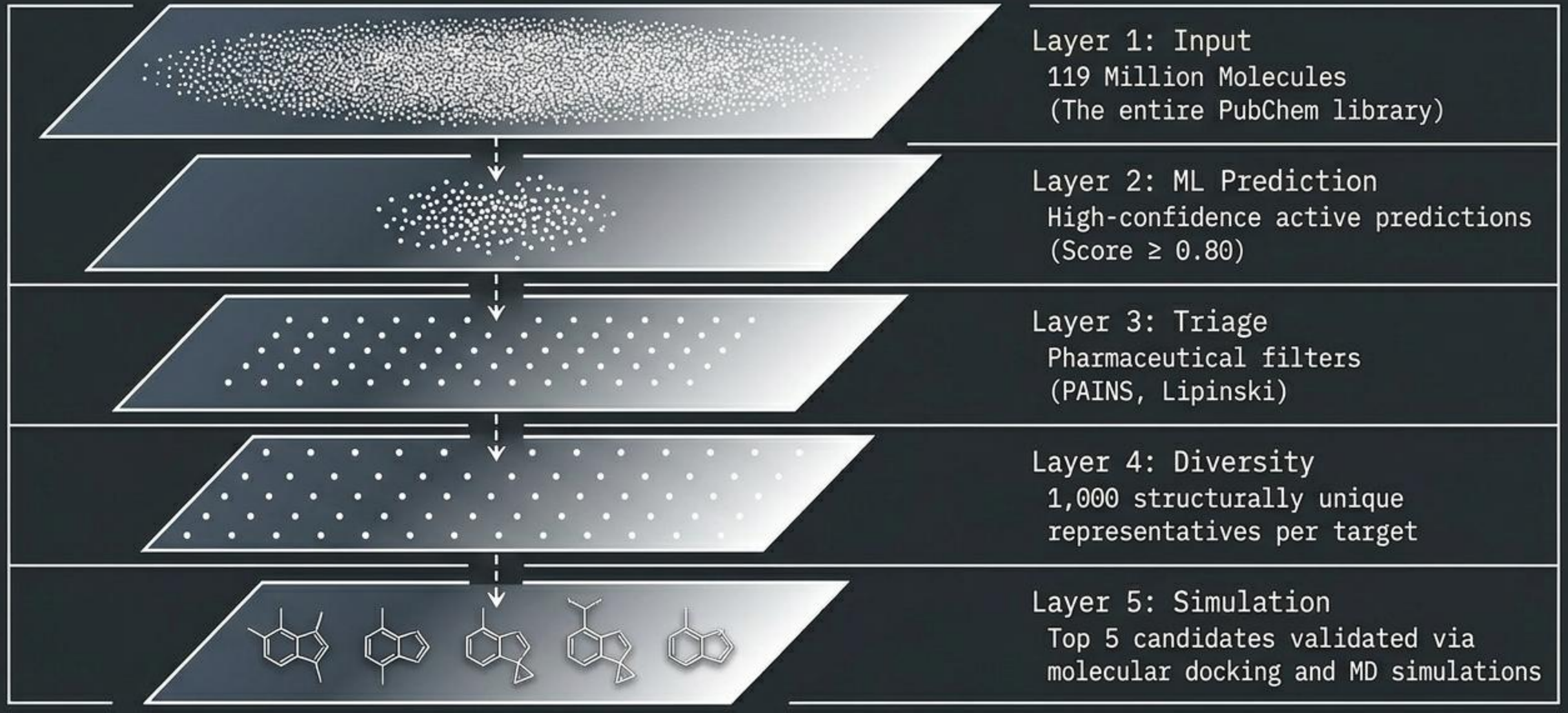
PDB ID: 1JPW

Biological Role: Wnt signalling effector; driver of colorectal & other cancers

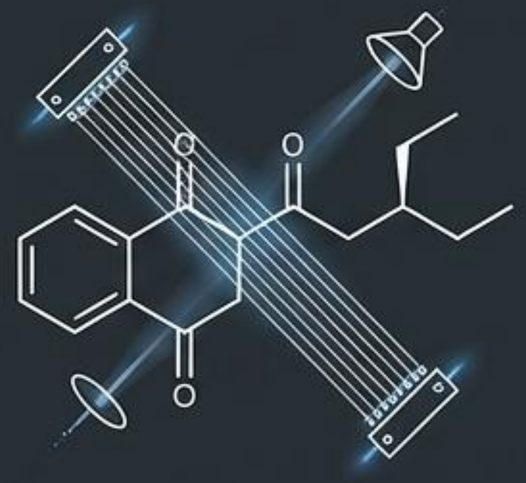
Raw Bioassay Files: ~80 CSV files (PubChem)

Curated Samples: 715 (72% active)

# The Scale-Down Architecture



# Translating Bioactivity into Binary Vectors



### 1. Data Curation

Aggregate PubChem BioAssay outcomes into clean datasets.

### 2. Molecular Fingerprinting

Convert SMILES strings to ECFP4 (Morgan) fingerprints via RDKit (Radius = 2, 2048 bits).

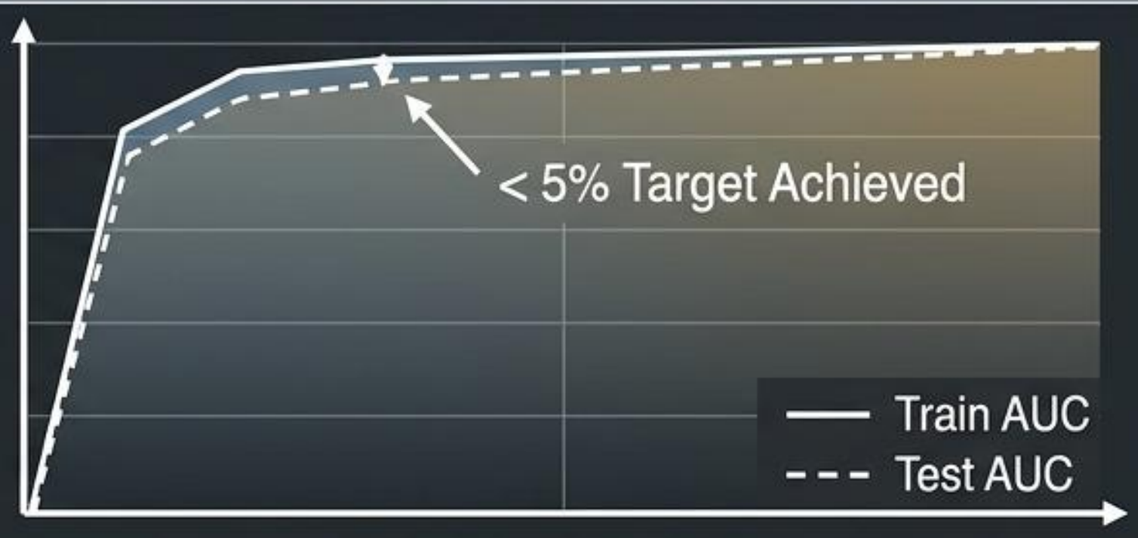
### 3. Redundancy Elimination

Drop highly correlated features (Pearson r > 0.90) prior to model exposure.

**Output Stats:** Initial feature space reduced to 1,597 bits for MYC and 1,471 bits for CTNNB1, ready for rigorous selection.

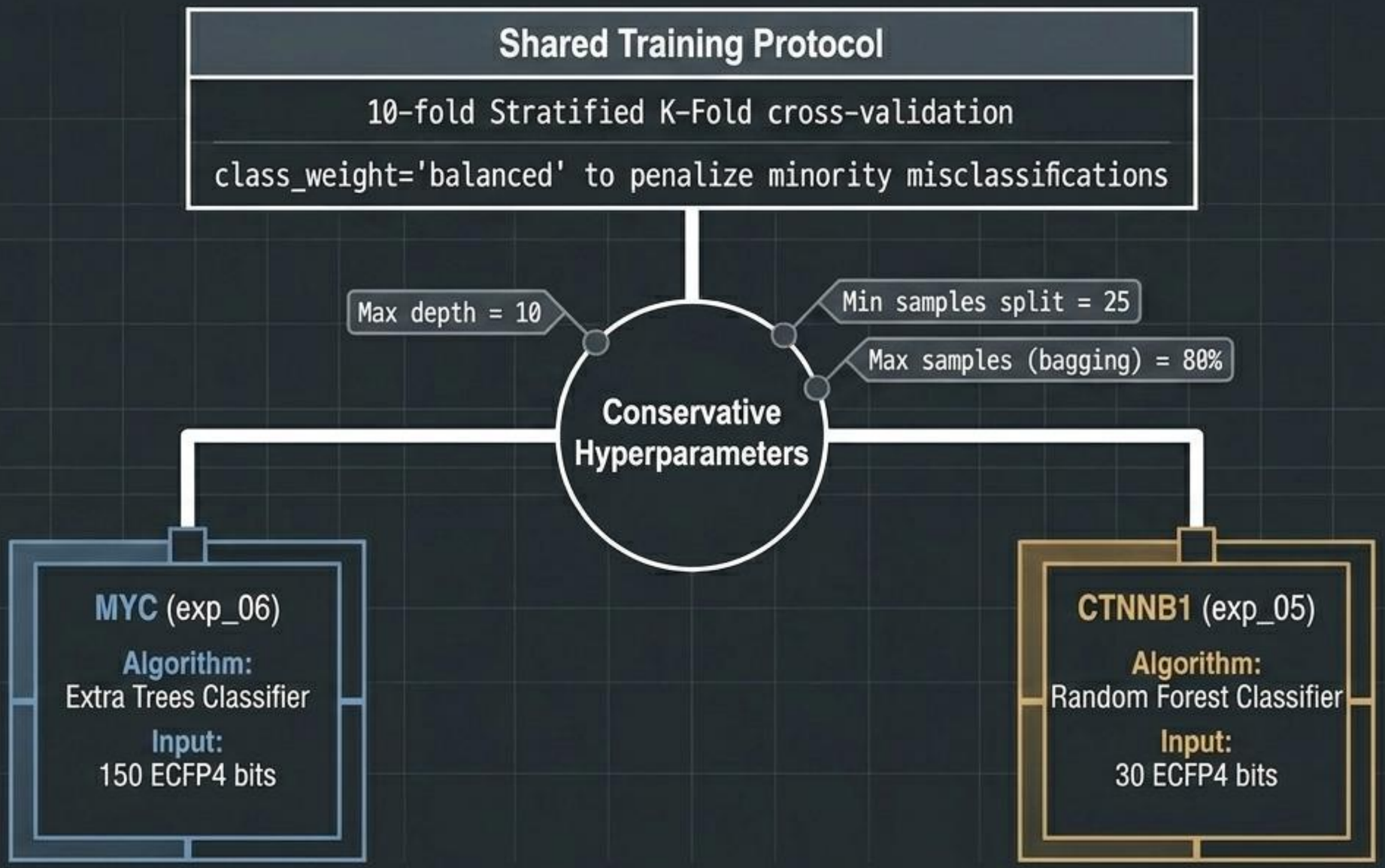
# Leakage-Free Feature Selection

**Methodological Rule:** Feature importance calculated strictly on the 80% training split. No test data exposure.

| Previous Approach (exp_03) | Robust Approach (Optimal K) |
|---|---|
| |   |
| ~315 features | ~150 features for MYC / ~30 features for CTNNB1 |
| **MYC**: 150 features (–52% vs exp_03)<br>`Train AUC: 1.000 / Test AUC: 0.952`<br>`Overfitting Gap: 4.75%` | **CTNNB1**: 30 features (–90% vs exp_03)<br>`Train AUC: 0.987 / Test AUC: 0.939`<br>`Overfitting Gap: 4.74%` |

**Impact:** Dramatically reduced model variance while increasing screening velocity by **up to 90%.**

# Anti-Overfitting Model Architecture

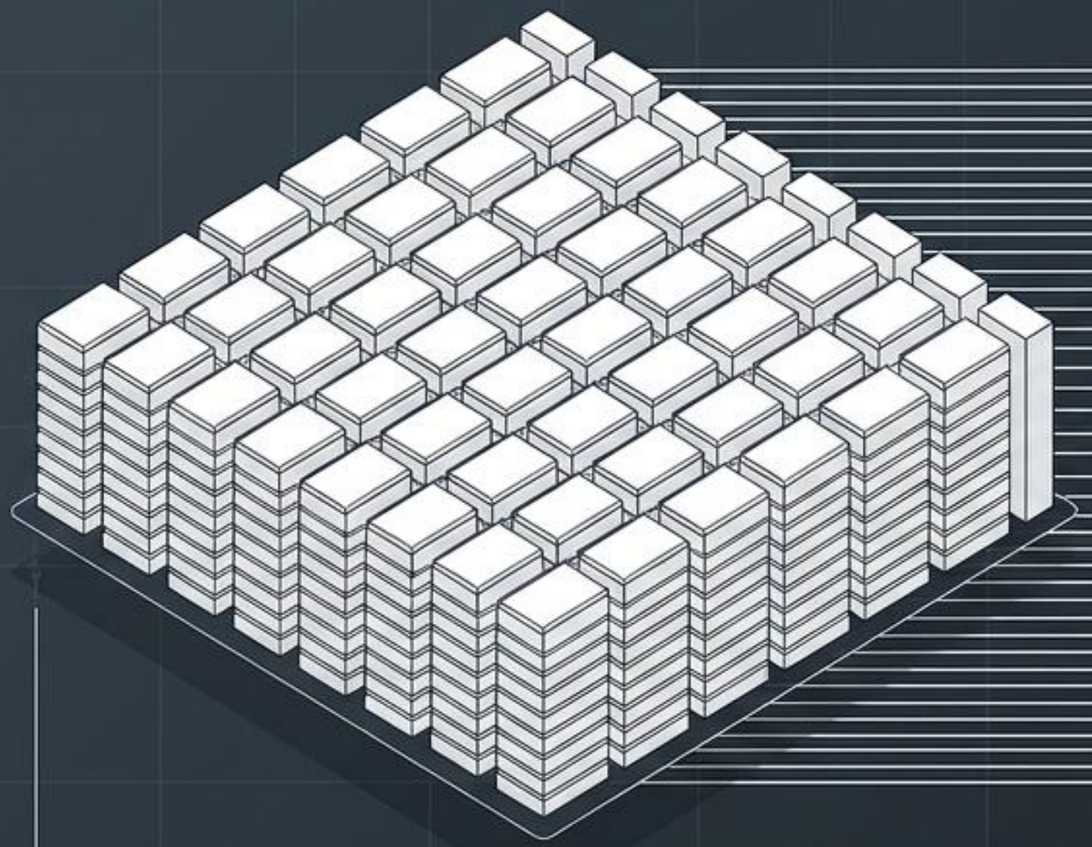


# Predictive Performance Validation

| Metric | MYC (Extra Trees) | CTNNB1 (Random Forest) |
|---|---|---|
| Features | 150 ECFP4 bits | 30 ECFP4 bits |
| Test AUC | 0.952 | 0.939 |
| Test F1 | 0.929 | 0.909 |
| Test Precision | 0.919 | 0.947 |
| Test Recall | 0.938 | 0.874 |
| Overfitting Gap | 4.75% | 4.74% |

Conclusion: Both .pkl models achieve >0.93 AUC with high precision, ensuring minimal false positives during mass screening.

# Massive-Scale Virtual Screening




The Activation Gate (ML Score ≥ 0.80)

Computational Execution: Processed in 60 parallel chunks via joblib with real-time ECFP4-2048 fingerprint generation.

# Pharmaceutical Triage and Diversity Clustering

**Filter 1: Structural Quality**

Applied PAINS and Brenk filters to eliminate pan-assay interference and false positives.

**Filter 2: Drug-Likeness**

Enforced Lipinski's Rule of Five (≤2 violations) and Veber rules for oral bioavailability.

**Filter 3: Chemical Space Coverage**

Applied Tanimoto Diversity Clustering (cutoff = 0.60) to avoid selecting minor scaffold variations.

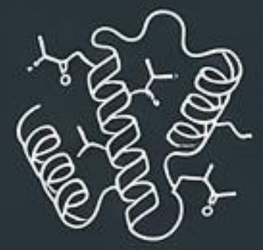
**Final Yield: Exactly 1,000 structurally unique, highly viable candidate representatives per target.**

# Dependency-Free Automated Docking Toolchain

Fully automated pipeline targeting the MYC–MAX protein-protein interaction interface (6G6K).

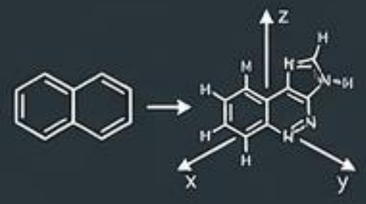
## Biopython

Automated protein cleaning and PDBQT receptor isolation.



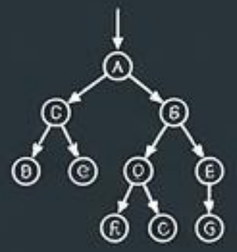
## RDKit

Conversion of 2D SMILES into 3D conformers with precise hydrogen addition.




## Meeko

Ligand PDBQT preparation (assigning atom types and torsion trees).



## AutoDock Vina

Execution via local pre-compiled binaries (no sudo/GUI required).

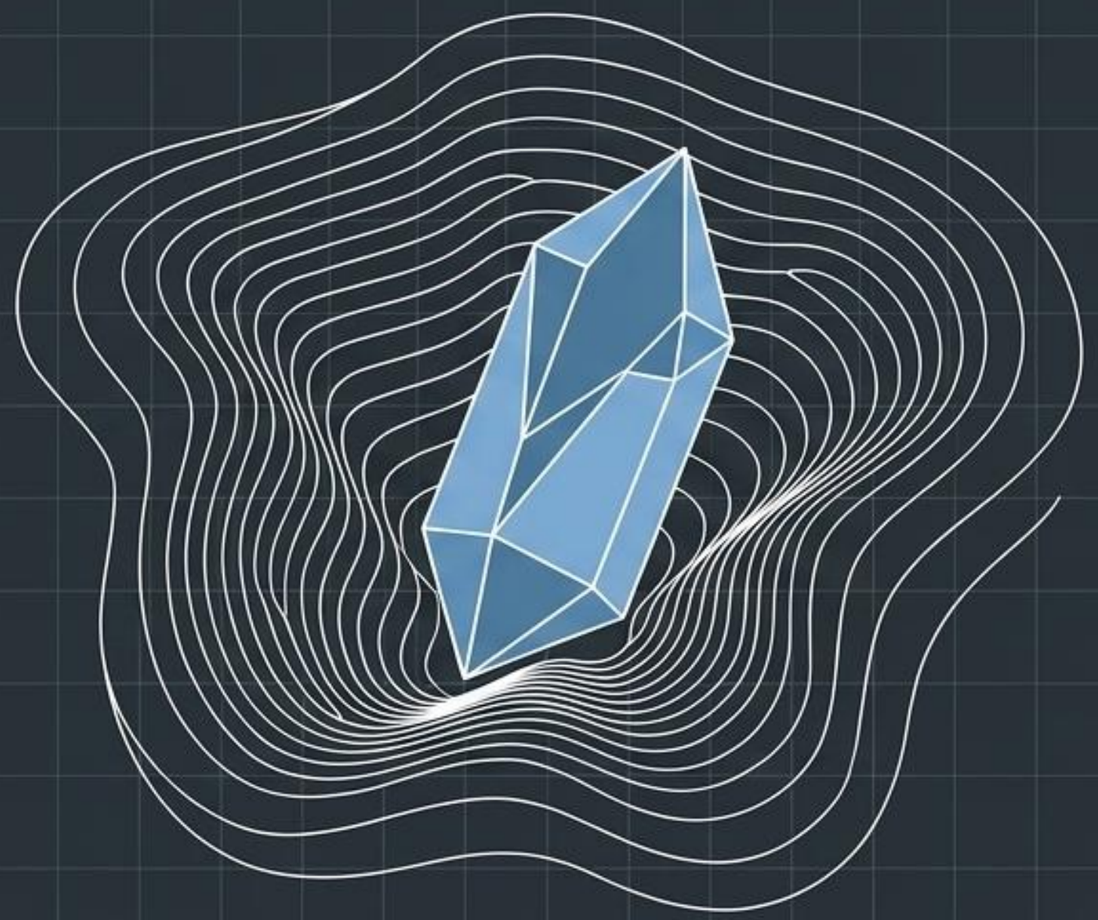
Advantage: Seamless progression from ML candidate directly into 3D binding simulation without manual file manipulation.

# Target Binding Results: MYC (PDB: 6G6K)

## Top 5 Cluster Representatives (AutoDock Vina)

| PubChem CID | Binding Affinity | # Poses |
|---|---|---|
| CID 124117970 | -7.56 kcal/mol | 9 Poses |
| CID 137355896 | -7.42 kcal/mol | 9 Poses |
| CID 155548927 | -6.98 kcal/mol | 9 Poses |
| CID 46206285 | -6.76 kcal/mol | 9 Poses |
| CID 91515241 | -6.42 kcal/mol | 9 Poses |

**Insight:** All final simulated candidates successfully demonstrate strong binding affinities within the −6.4 to −7.6 kcal/mol range at the critical MYC–MAX interface.



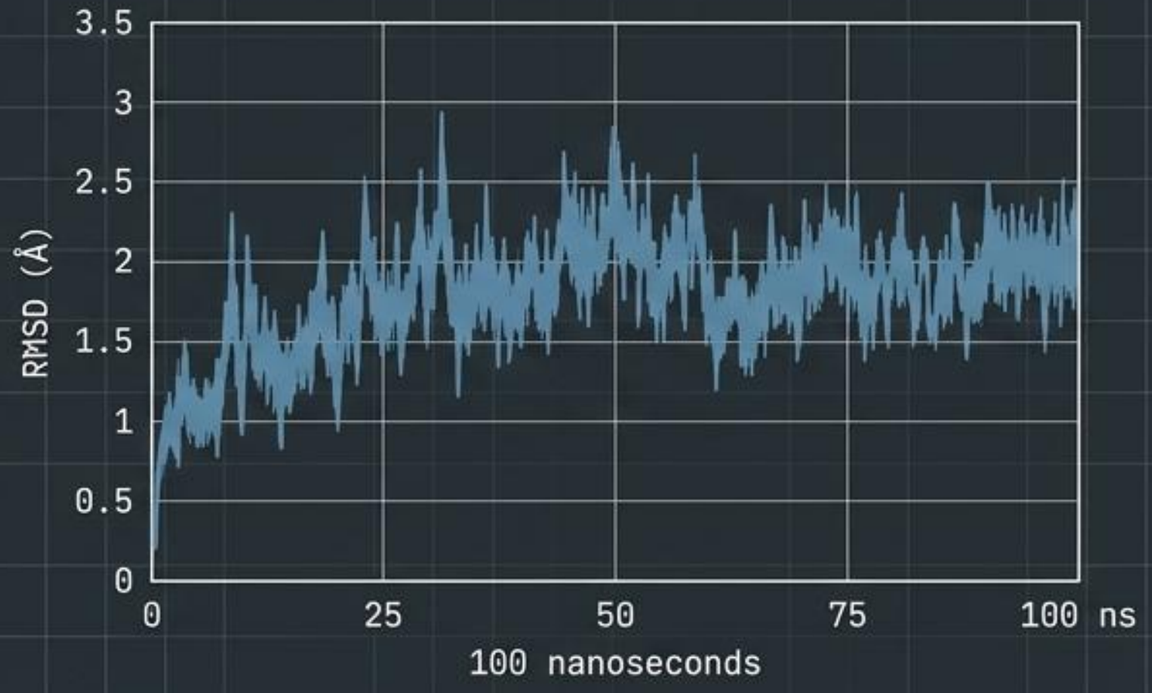
# Dynamic Validation via MD Simulations

Outcome: MD trajectories will dictate the final, definitive ranking of the candidates.

## Limitation: Static Binding (AutoDock Vina)

AutoDock Vina provides a static snapshot of binding affinity.

## Current Execution: Dynamic Stability (MD Simulations)




Objective: To measure dynamic binding stability, track RMSD over time, and confirm the ligand remains securely locked in the MYC–MAX interface under physiological conditions.

# Interactive Deployment: The Web Portal

### Protein Selector

Tab-based navigation between MYC and CTNNB1 datasets.

### Export Ready

Direct CSV export functionality to enable immediate lab testing.

### Interactive Analytics

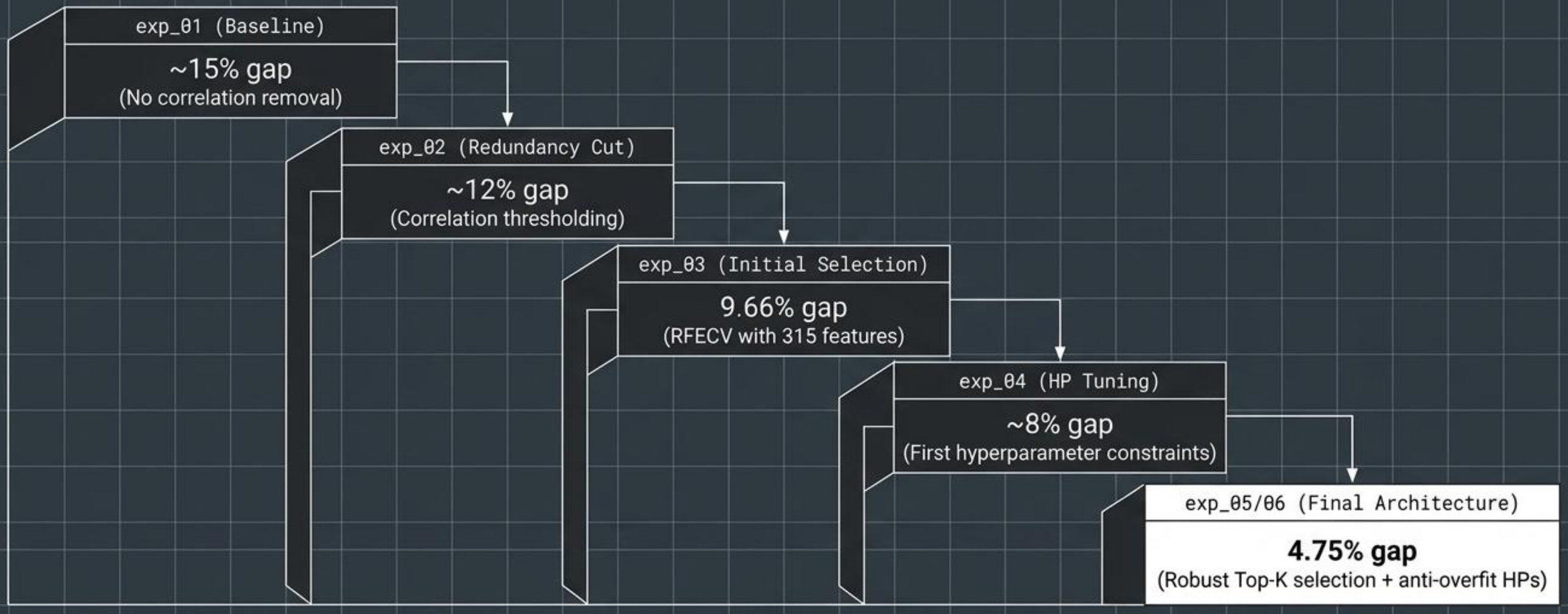
Plotly scatter plots comparing ML scores against molecular weight, logP, and TPSA.

### Candidate Triage

Sortable tables featuring CIDs, SMILES, and Lipinski flags.

**Platform**: Streamlit-based visualization dashboard built with an Inter-inspired academic dark-mode UI.

# Methodological Evolution & Impact




**Synthesis:** By treating feature selection and generalization as the primary engineering constraints, the pipeline evolved from a standard ML script into a rigorous, leakage-free discovery engine capable of scanning 119M compounds safely.

NotebookLM

# Strategic Trajectory and Next Steps

**Project Objective Maintained:**
Delivering open, reproducible, high-precision hit generation for hard-to-drug oncogenic targets.

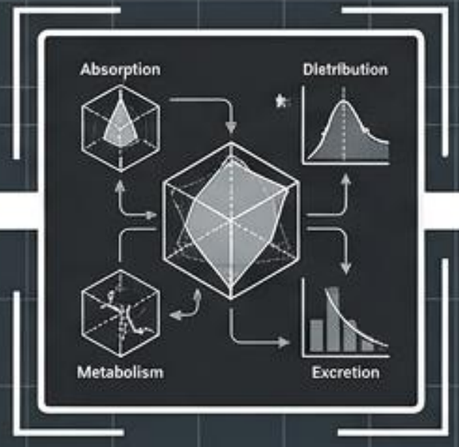
## 1. Trajectory Integration

Complete 100 ns MD simulations and merge dynamic stability metrics into the final MYC candidate rankings.

## 2. Target Expansion

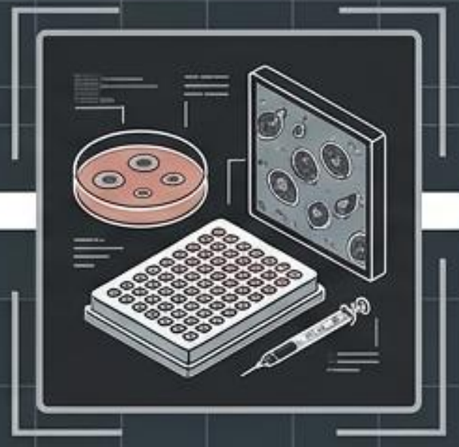
Execute the automated AutoDock Vina pipeline for the top 5 CTNNB1 cluster representatives.

## 3. Pharmacokinetic Profiling




Conduct advanced ADMET computational profiling for the finalized candidate lists.

## 4. Experimental Validation



Transition the top 10 computationally validated compounds toward in-vitro biological testing.

NotebookLM

THANKS ..